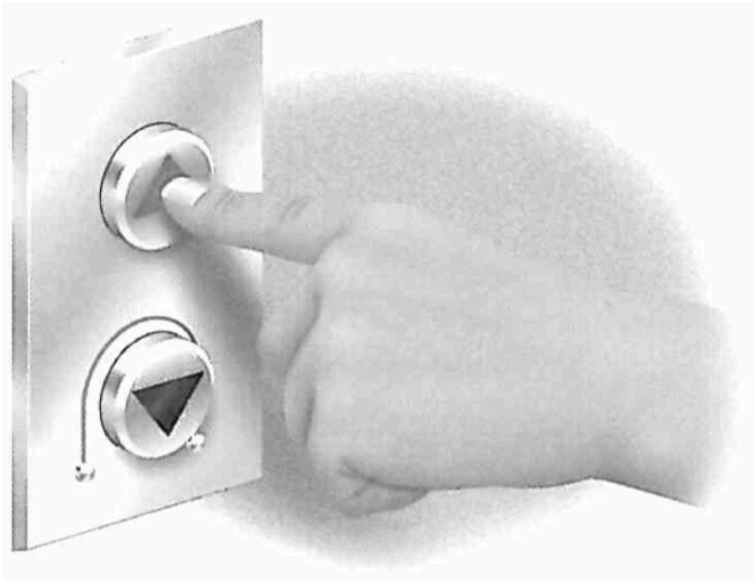
Inductance, Capacitance, and Mutual Inductance

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Structure

- The Inductor
- The Capacitor
- Series-Parallel Combinations of Inductance and Capacitance
- Mutual Inductance
- A Closer Look at Mutual Inductance

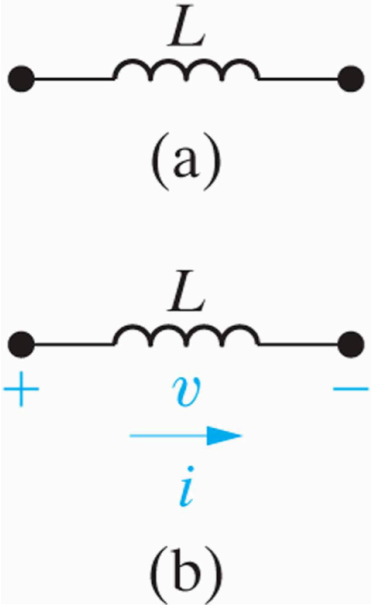
Proximity Switches



Sometimes designers prefer to use switches without moving parts, to increase the *safety*, *reliability*, *convenience*, or *novelty* of their products. Such switches are called proximity switches.

Proximity switches can employ a variety of sensor technologies. For example, some *elevator doors* stay open whenever a light beam is obstructed. We will introduce the design of a capacitive touch -sensitive switch at the end of this chapter.

The Inductor



Inductance is the circuit parameter used to describe an inductor. Inductance is *symbolized by the letter L*, is *measured in henrys (H)*, and is *represented graphically as a coiled wire*—a reminder that inductance is a consequence of a conductor linking a magnetic field.

Assigning the reference direction of the current in *the direction of the voltage drop across the terminals of the inductor*, as shown in (b), yields

$$v = L\frac{di}{dt}$$

Two Important Observations

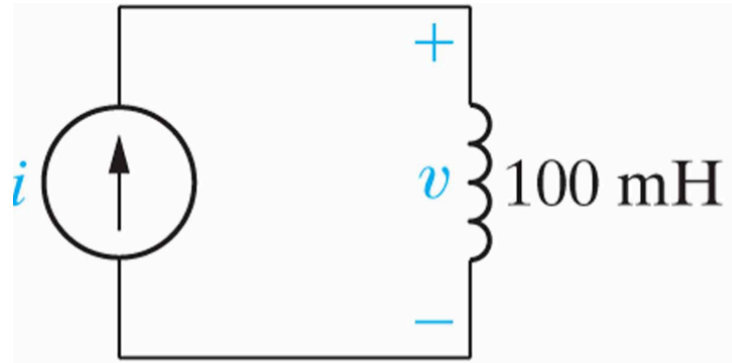
a) If the current is constant, the voltage across the ideal inductor is zero. Thus *the inductor behaves as a short circuit in the presence of a constant, or dc, current*.

b) Current cannot change instantaneously in an inductor; that is, *the current cannot change by a finite amount in zero time.*

- When someone opens the switch on an inductive circuit in an actual system, the current initially continues to flow in the air across the switch, a phenomenon called *arcing.*
- Arcing must be controlled to prevent equipment damage!

Example #1

- The independent current source in the circuit generates zero current for $t < 0$ and a pulse $10te^{-5t}$A, for $t > 0$.



$$i = 0, \qquad t < 0$$

$$i = 10te^{-5t}\text{A}, \qquad t > 0$$

a) Sketch the current waveform.

b) At what instant of time is the current maximum?

c) Express the voltage across the terminals of the 100 mH inductor as a function of time.

d) Sketch the voltage waveform.

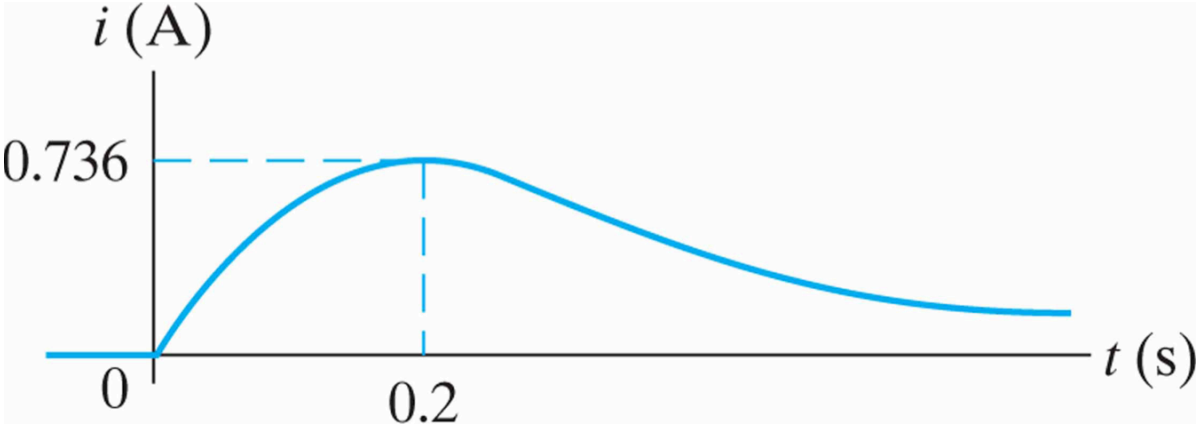
e) Are the voltage and the current at a maximum at the same time?

f) At what instant of time does the voltage change polarity?

g) Is there ever an instantaneous change in voltage across the inductor? If so, at what time?

Solution for Example #1

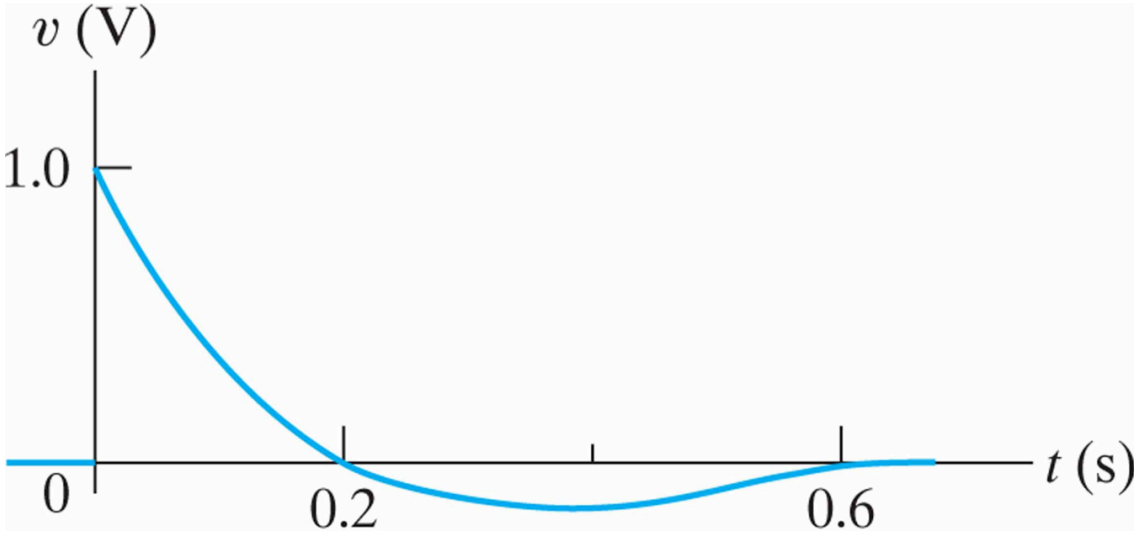
a)



b) $di/dt = 10(-5te^{-5t} + e^{-5t}) = 10e^{-5t}$
$(1-5t)$ A/s; $di/dt = 0$ when $t = \frac{1}{5}$ s.

c) $v = Ldi/dt = (0.1)10e^{-5t}(1 - 5t) = e^{-5t}$
$(1-5t)$ V, $t > 0$; $v = 0$, $t < 0$.

d)



e) No; the voltage is proportional to di/dt, not i.

f) At 0.2 s, which corresponds to the moment when di/dt is passing through zero and changing sign.

g) Yes, at t - 0. Note that the voltage can change instantaneously across the terminals of an inductor.

Current in an Inductor in Terms of the Voltage Across the Inductor

$$v = L\frac{di}{dt} \Rightarrow v\,dt = L\,di \xrightarrow{\text{Integrate}} L\int_{i(t_0)}^{i(t)} dx = \int_{t_0}^{t} v\,d\tau$$

$$i(t) = \frac{1}{L}\int_{t_0}^{t} v\,d\tau + i(t_0)$$

$$t_0 = 0 \Rightarrow i(t) = \frac{1}{L}\int_{0}^{t} v\,d\tau + i(0)$$

Power and Energy in the Inductor

$$p = vi \Rightarrow p = Li\frac{di}{dt} \Rightarrow p = v\left[\frac{1}{L}\int_{t_0}^{t} v\,d\tau + i(t_0)\right]$$

$$p = \frac{dw}{dt} = Li\frac{di}{dt} \Rightarrow dw = Li\,di$$

$$\int_0^w dx = L\int_0^i y\,dy \Rightarrow w = \frac{1}{2}Li^2$$

Example #2

a) For Example #1, Plot i, v, p, and w versus time. Line up the plots vertically to allow easy assessment of each variable's behavior.

b) In what time interval is energy being stored in the inductor?

c) In what time interval is energy being extracted from the inductor?

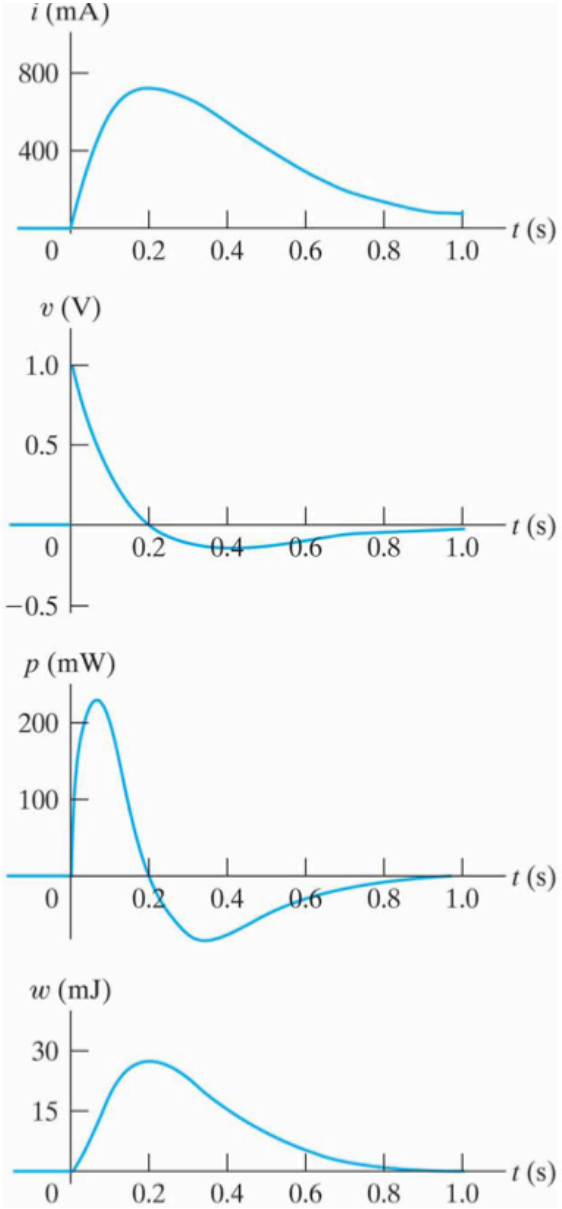
d) What is the maximum energy stored in the inductor?

e) Evaluate the integrals and comment on their significance.

$$\int_0^{0.2} p\,dt \qquad \text{and} \qquad \int_{0.2}^{\infty} p\,dt$$

a)

Solution for Example #2



b) An increasing energy curve indicates that energy is being stored. Thus energy is being stored in the time interval 0 to 0.2 s. *Note that this corresponds to the interval when $p > 0$.*

c) A decreasing energy curve indicates that energy is being extracted. Thus energy is being extracted in the time interval 0.2 s to ∞. *Note that this corresponds to the interval when $p < 0$.*

d) *Energy is at a maximum when current is at a maximum*; glancing at the graphs confirms this. From Example #1, maximum current = 0.736 A. Therefore, $w_{max} = 27.07$ mJ.

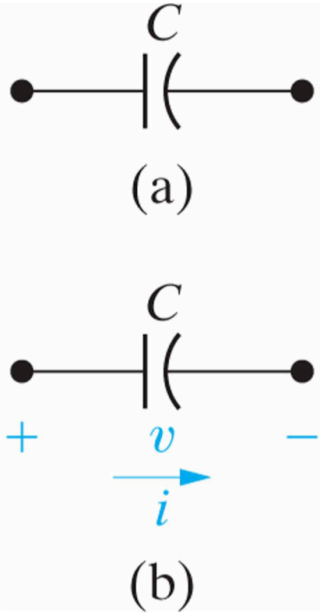
e) From Example #1, we have $i(t) = 10te^{-5t}$ A and $L = 0.1$ H, therefore,

$$\int_0^{0.2} p\,dt = w(0.2) - w(0) = L/2 \times [i(0.2)^2 - i(0)^2] = 0.2e^{-2} = 27.07 \text{ mJ}$$

$$\int_{0.2}^{\infty} p\,dt = w(\infty) - w(0.2) = L/2 \times [i(\infty)^2 - i(0.2)^2] = -0.2e^{-2} = -27.07 \text{ mJ}$$

Based on the definition of p, the area under the plot of p versus t represents the energy expended over the interval of integration. Hence the integration of the power between 0 and 0.2 s represents the energy stored in the inductor during this time interval. The integral of p over the interval 0.2 s - ∞ is the energy extracted. *Note that in this time interval, all the energy originally stored is removed; that is, after the current peak has passed, no energy is stored in the inductor.*

The Capacitor



The circuit parameter of capacitance is represented by the letter C, **is** measured in farads (F), and is symbolized graphically by two short parallel conductive plates. Because the farad is an extremely large quantity of capacitance, practical capacitor values usually lie in the picofarad (pF) to microfarad (mF) range.

The graphic symbol for a capacitor is a reminder that *capacitance occurs whenever electrical conductors are separated by a dielectric, or insulating, material*. This condition implies that *electric charge is not transported through the capacitor*.

Formulas

$$i = C\frac{dv}{dt} \Rightarrow i\,dt = C\,dv \xrightarrow{\text{Integrate}} \int_{v(t_0)}^{v(t)} dx = \frac{1}{C}\int_{t_0}^{t} i\,d\tau$$

$$v(t) = \frac{1}{C}\int_{t_0}^{t} i\,d\tau + v(t_0) \xrightarrow{t_0 = 0} v(t) = \frac{1}{C}\int_{0}^{t} i\,d\tau + v(0)$$

Power and Energy in the Capacitor

$$p = vi = Cv\frac{dv}{dt} \quad \Rightarrow \quad p = i\left[\frac{1}{C}\int_{t_0}^{t} i\,d\tau + v(t_0)\right]$$

$$dw = Cv\,dv \quad \Rightarrow \quad \int_0^w dx = C\int_0^v y\,dy \quad \Rightarrow \quad w = \frac{1}{2}Cv^2$$

Example #3

The voltage pulse described by the following equations is impressed across the terminals of a $0.5\ \mu\text{F}$ capacitor:

$$v(t) = \begin{cases} 0, & t \leq 0\ \text{s}; \\ 4t\ \text{V}, & 0\ \text{s} \leq t \leq 1\ \text{s}; \\ 4e^{-(t-1)}\ \text{V}, & t \geq 1\ \text{s}. \end{cases}$$

a) Derive the expressions for the capacitor current, power, and energy.

b) Sketch the voltage, current, power, and energy as functions of time. Line up the plots vertically.

c) Specify the interval of time when energy is being stored in the capacitor.

d) Specify the interval of time when energy is being delivered by the capacitor.

e) Evaluate the integrals and comment on their significance.

$$\int_0^1 p\,dt \qquad \text{and} \qquad \int_1^\infty p\,dt$$

Solution for Example #3

a)

$$i = C\frac{dv}{dt} \Rightarrow i = \begin{cases} (0.5 \times 10^{-6})(0) = 0, & t < 0\text{s}; \\ (0.5 \times 10^{-6})(4) = 2\,\mu\text{A}, & 0\,\text{s} < t < 1\,\text{s}; \\ (0.5 \times 10^{-6})(-4e^{-(t-1)}) = -2e^{-(t-1)}\,\mu\text{A}, & t > 1\,\text{s}. \end{cases}$$

$$p = vi = Cv\frac{dv}{dt} \Rightarrow p = \begin{cases} 0, & t \leq 0\,\text{s}; \\ (4t)(2) = 8t\,\mu\text{W}, & 0\,\text{s} \leq t < 1\,\text{s}; \\ (4e^{-(t-1)})(-2e^{-(t-1)}) = -8e^{-2(t-1)}\,\mu\text{W}, & t > 1\,\text{s}. \end{cases}$$

$$w = \frac{1}{2}Cv^2 \Rightarrow w = \begin{cases} 0 & t \leq 0\,\text{s}; \\ \frac{1}{2}(0.5)16t^2 = 4t^2\mu\text{J}, & 0\,\text{s} \leq t \leq 1\,\text{s}; \\ \frac{1}{2}(0.5)16e^{-2(t-1)} = 4e^{-2(t-1)}\,\mu\text{J}, & t \geq 1\,\text{s}. \end{cases}$$

a) Figure at right shows the voltage, current, power, and energy as functions of time.

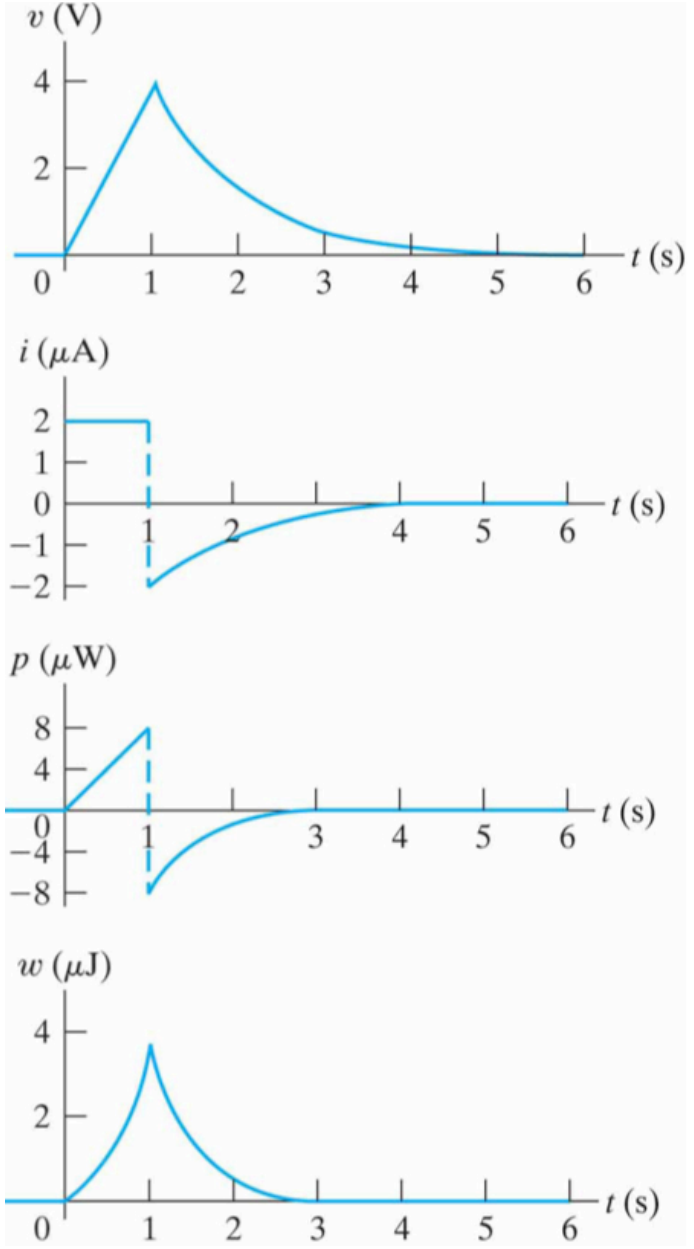
b) Energy is being stored in the capacitor whenever the power is positive. Hence energy is being stored in the interval 0-1 s.

c) Energy is being delivered by the capacitor whenever the power is negative. Thus energy is being delivered for all / greater than 1 s.

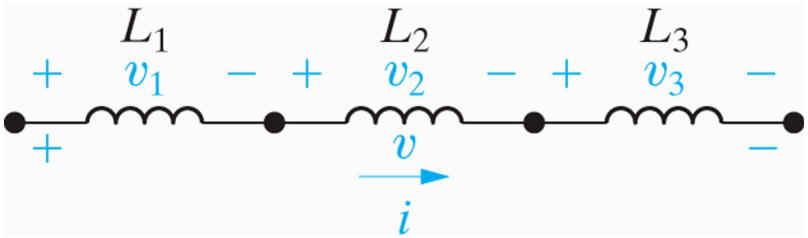
d) The integral of $p\,dt$ is the energy associated with the time interval corresponding to the limits on the integral. Thus the first integral represents the energy stored in the capacitor between 0 and 1 s, whereas the second integral represents the energy returned, or delivered, by the capacitor in the interval 1 s to ∞:

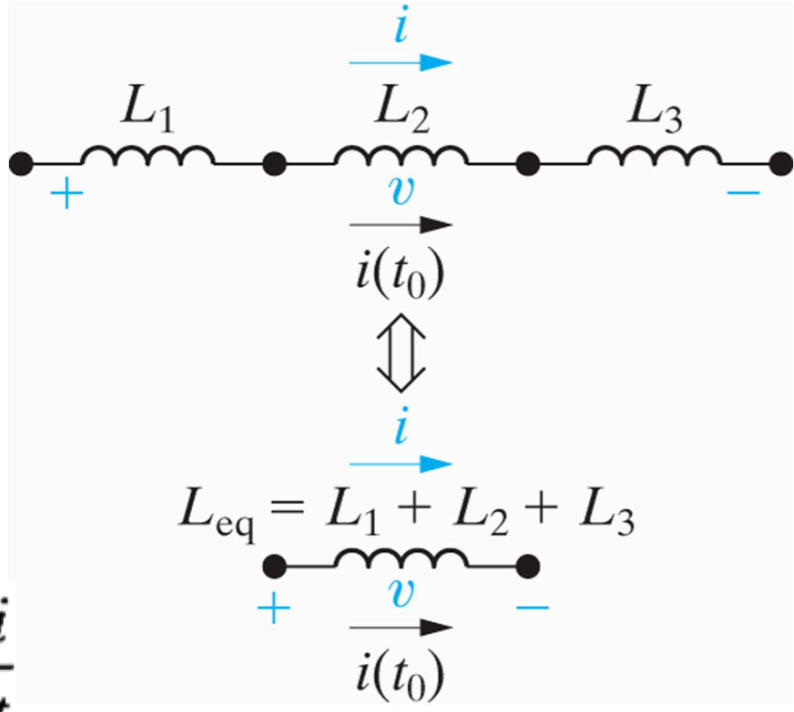
$$\int_0^1 p\,dt = \int_0^1 8t\,dt = 4t^2\Big|_0^1 = 4\ \mu\text{J}$$

$$\int_1^\infty p\,dt = \int_1^\infty (-8e^{-2(t-1)})dt = (-8)\frac{e^{-2(t-1)}}{-2}\Big|_1^\infty = -4\ \mu\text{J}$$



Series Combination of Inductance





$$v_1 = L_1 \frac{di}{dt} \qquad v_2 = L_2 \frac{di}{dt} \qquad v_3 = L_3 \frac{di}{dt}$$

$$v = v_1 + v_2 + v_3 = (L_1 + L_2 + L_3)\frac{di}{dt}$$

$$L_{eq} = L_1 + L_2 + L_3 + \cdots + L_n$$

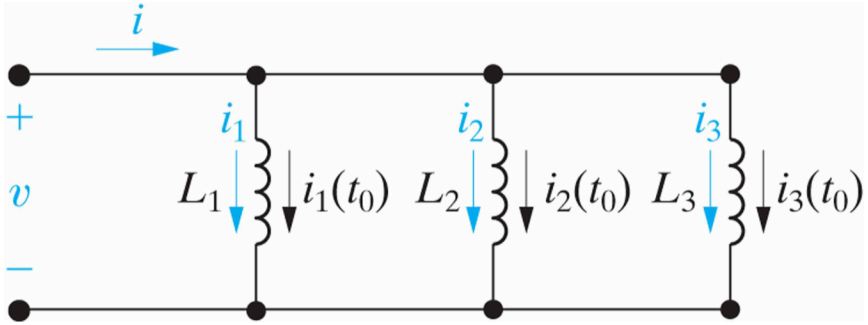
Parallel Combination of Inductance

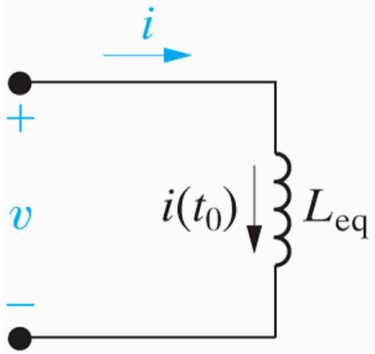
$$i_1 = \frac{1}{L_1}\int_{t_0}^{t} v\,d\tau + i_1(t_0)$$

$$i_2 = \frac{1}{L_2}\int_{t_0}^{t} v\,d\tau + i_2(t_0)$$

$$i_3 = \frac{1}{L_3}\int_{t_0}^{t} v\,d\tau + i_3(t_0)$$

$$i = i_1 + i_2 + i_3$$



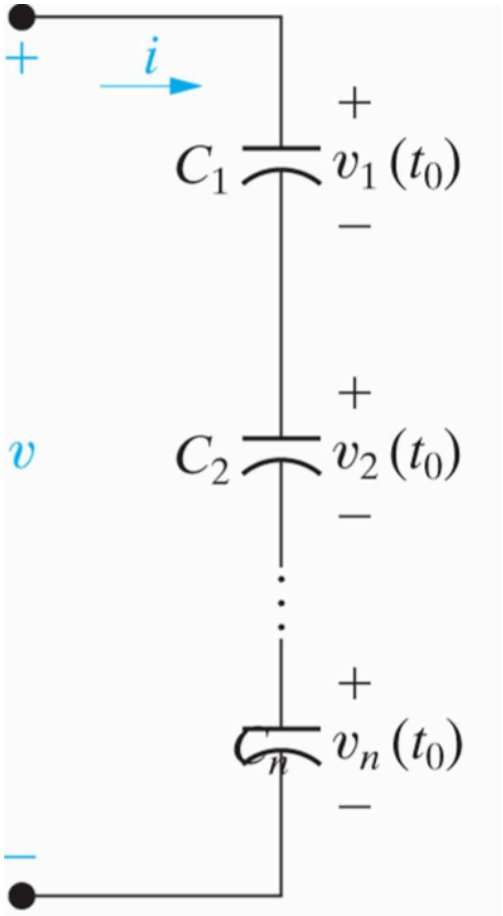


$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3}$$

$$i(t_0) = i_1(t_0) + i_2(t_0) + i_3(t_0)$$

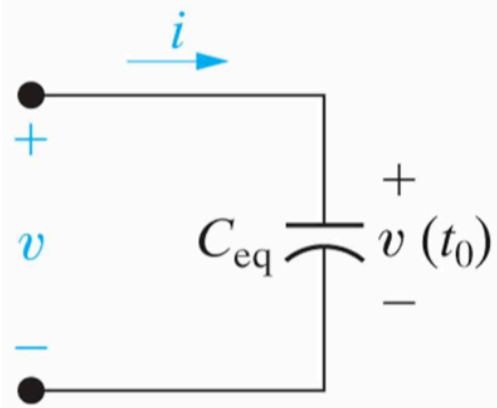
$$i = \left(\frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3}\right)\int_{t_0}^{t} v\,d\tau + i_1(t_0) + i_2(t_0) + i_3(t_0)$$

Series Combination of Capacitance



$$v(t) = \frac{1}{C}\int_0^t i\,d\tau + v(0)$$

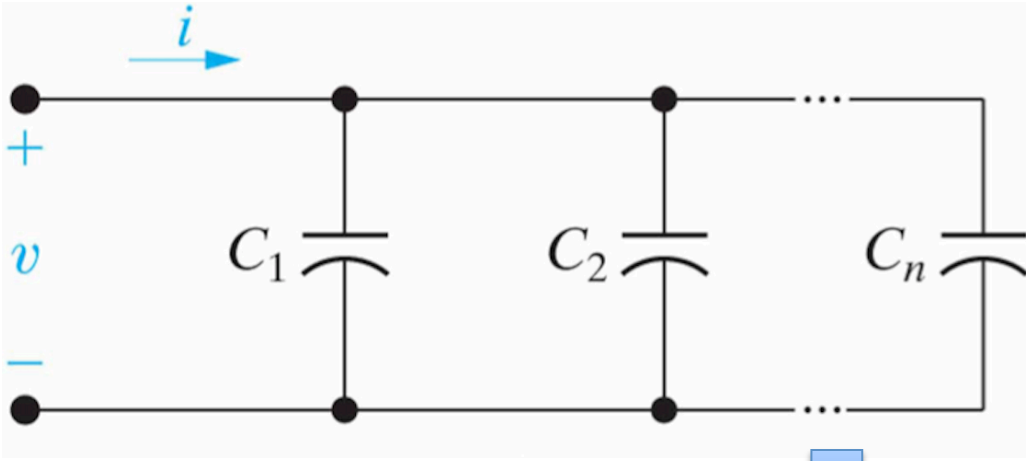




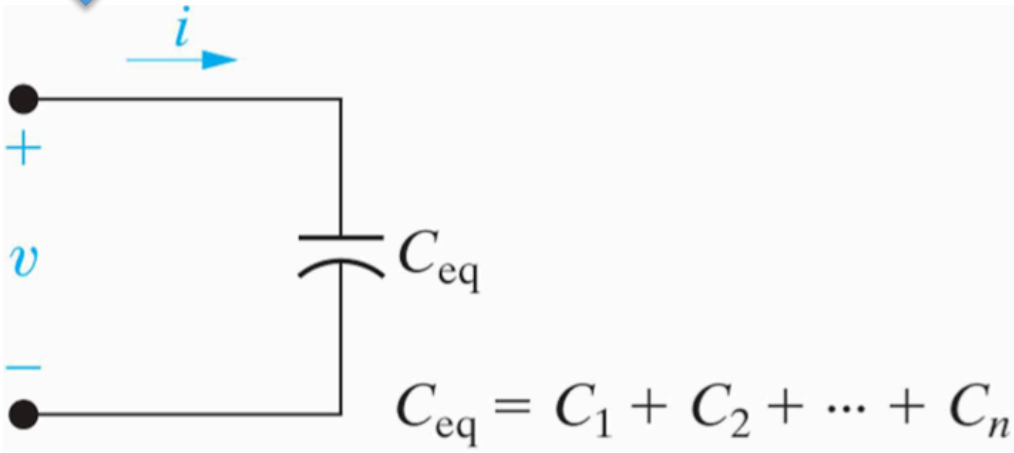
$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots + \frac{1}{C_n}$$

$$v(t_0) = v_1(t_0) + v_2(t_0) + \cdots + v_n(t_0)$$

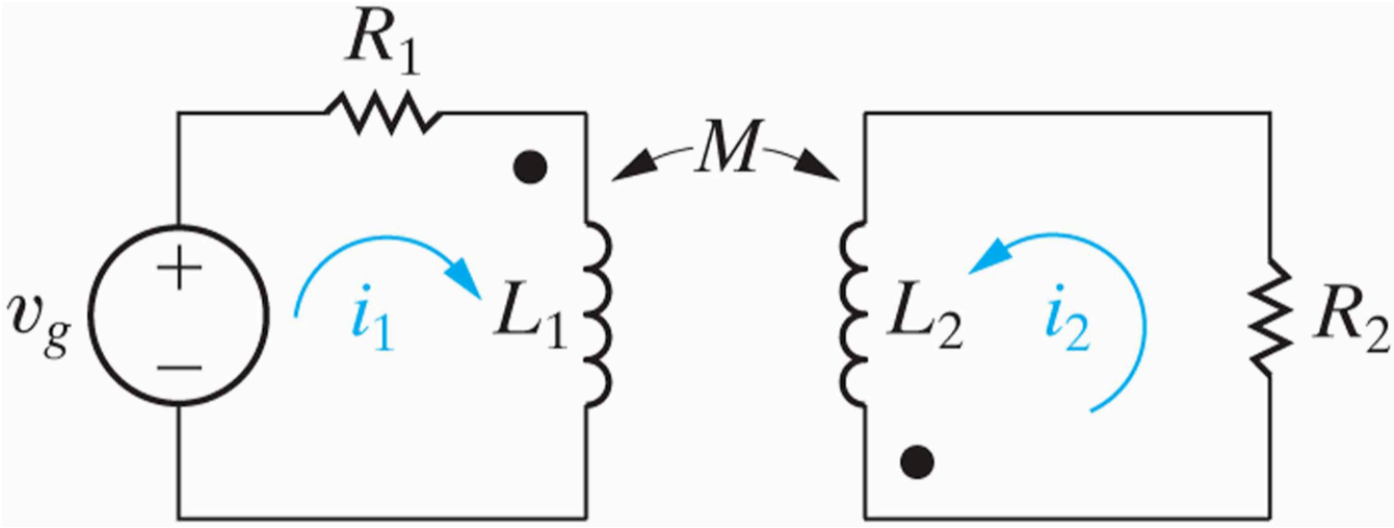
Parallel Combination of Capacitance



$$i = C\frac{dv}{dt}$$



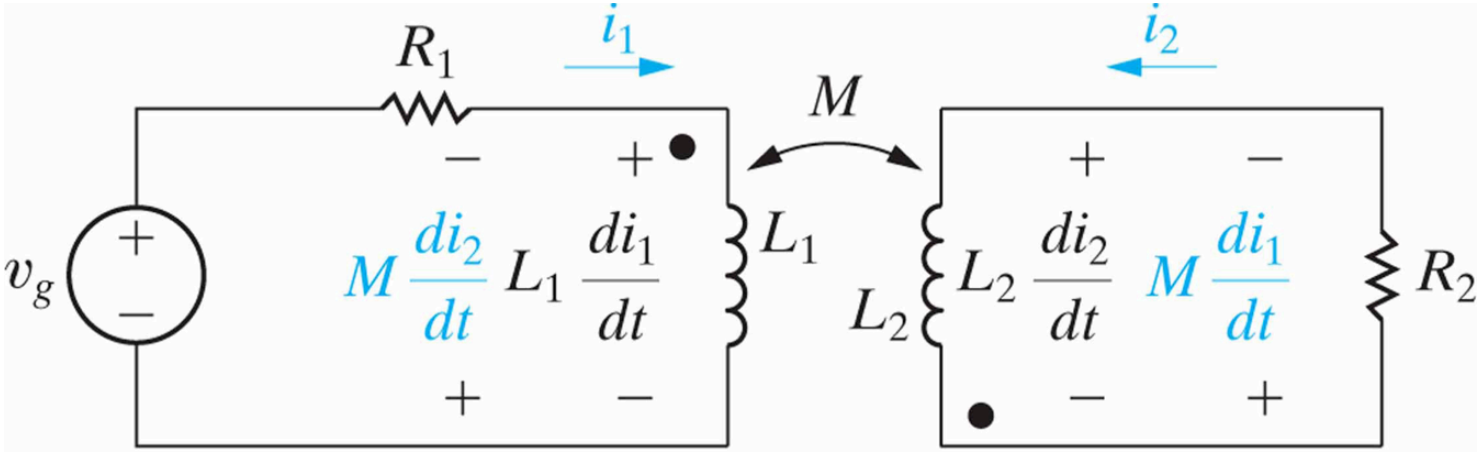
Mutual Inductance



L_1, L_2: self-inductances; M: mutual inductance

Dot Convention: When the reference direction for a current enters (leaves) the dotted terminal of a coil, the reference polarity of the voltage that it induces in the other coil is positive (negative) at its dotted terminal.

Equations



$$-v_g + i_1R_1 + L_1\frac{di_1}{dt} - M\frac{di_2}{dt} = 0$$

$$i_2R_2 + L_2\frac{di_2}{dt} - M\frac{di_1}{dt} = 0$$

Determining Dot Markings

a) Arbitrarily select one terminal—say, the D terminal—of one coil and mark it with a dot.

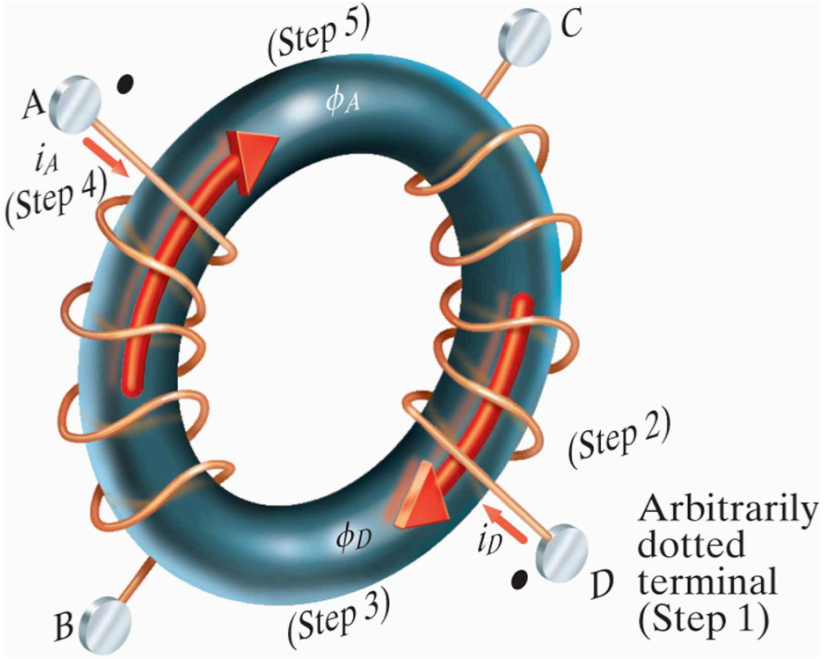
b) Assign a current into the dotted terminal and label it i_D.

c) Use the right-hand rule to determine the direction of the magnetic field established by i_D *inside* the coupled coils and label this field ϕ_D.

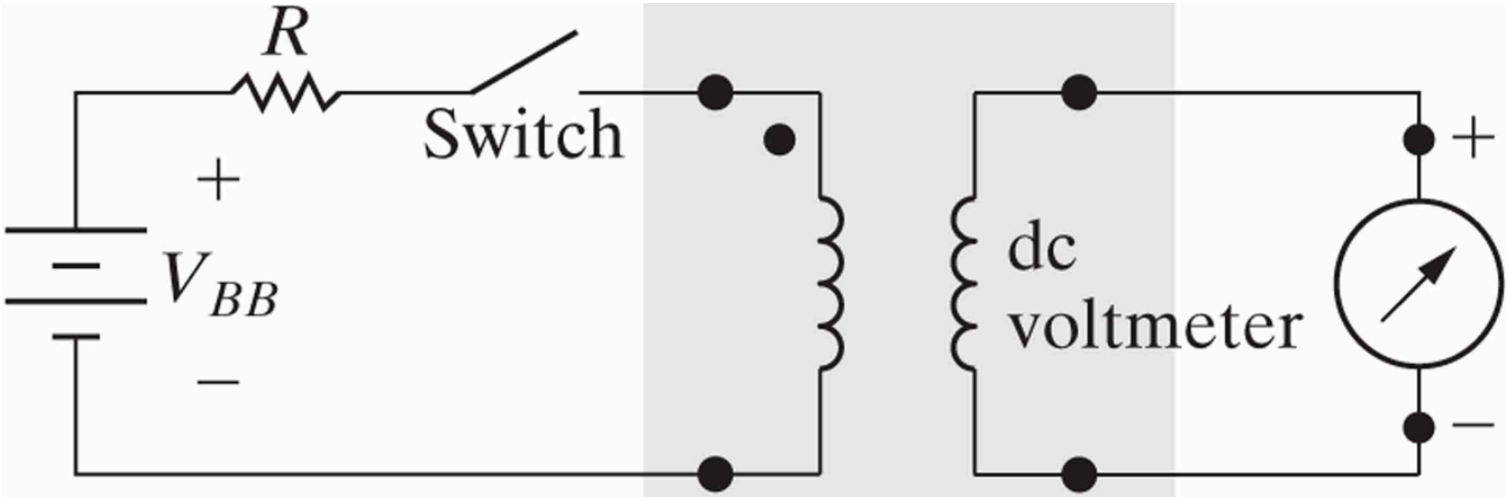
d) Arbitrarily pick one terminal of the second coil—say, terminal A—and assign a current into this terminal, showing the current as i_A.

e) Use the right-hand rule to determine the direction of the flux established by i_A *inside* the coupled coils and label this flux ϕ_A.

f) Compare the directions of the two fluxes ϕ_D and ϕ_A. If the fluxes have the same reference direction, place a dot on the terminal of the second coil where the test current (i_A) enters. (In the Figure, the fluxes ϕ_D and ϕ_A have the same reference direction, and therefore a dot goes on terminal A.) If the fluxes have different reference directions, place a dot on the terminal of the second coil where the test current leaves.



Experimental Method



When the switch is closed, the voltmeter deflection is observed.

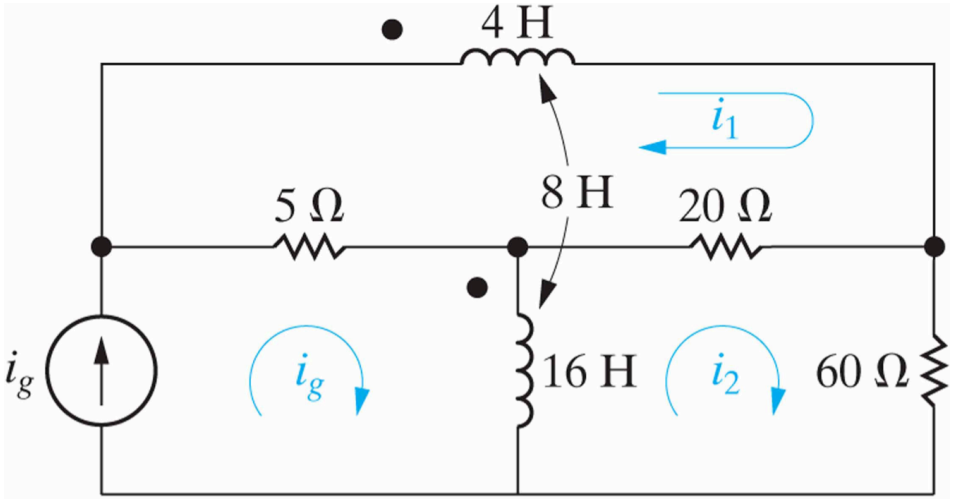
If the momentary deflection is upscale, the coil terminal connected to the positive terminal of the voltmeter receives the polarity mark.

If the deflection is downscale, the coil terminal connected to the negative terminal of the voltmeter receives the polarity mark.

Example #4

a) Write a set of mesh-current equations that describe the circuit in the Figure in terms of the currents i_1 and i_2.

b) Verify that if there is no energy stored in the circuit at $t = 0$ and if $i_g = 16 - 16e^{-5t}$ A, the solutions for i_1 and i_2 are



$$i_1 = 4 + 64e^{-5t} - 68e^{-4t} \text{ A}$$

$$i_2 = 1 - 52e^{-5t} + 51e^{-4t} \text{ A}$$

Solution for Example #4

a) $$4\frac{di_1}{dt} + 8\frac{d}{dt}(i_g - i_2) + 20(i_1 - i_2) + 5(i_1 - i_g) = 0$$

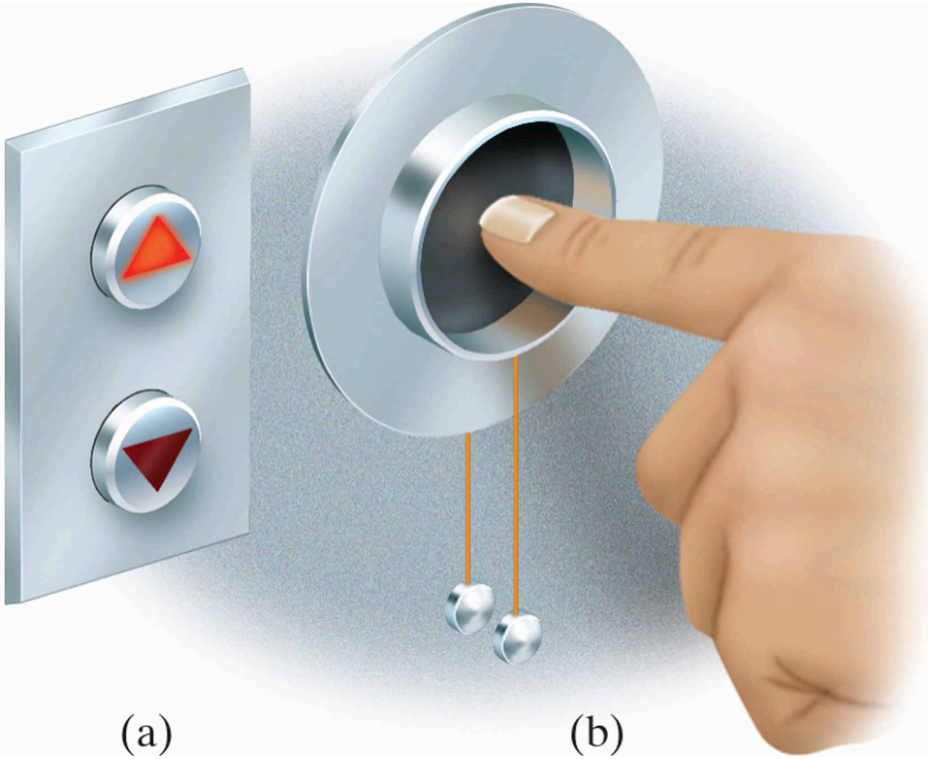
$$20(i_2 - i_1) + 60i_2 + 16\frac{d}{dt}(i_2 - i_g) - 8\frac{di_1}{dt} = 0$$

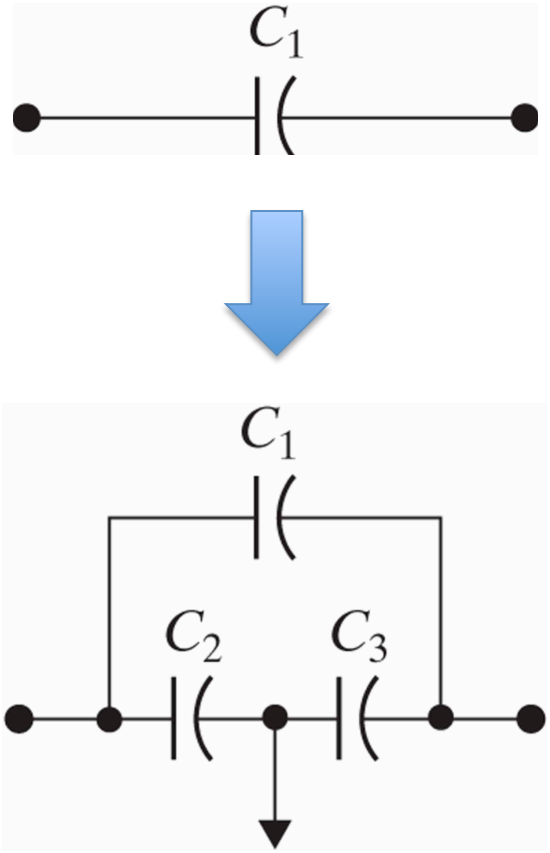
Substitute

b) $$i_1 = 4 + 64e^{-5t} - 68e^{-4t}\ \text{A}$$

$$i_2 = 1 - 52e^{-5t} + 51e^{-4t}\ \text{A}$$

Proximity Switches





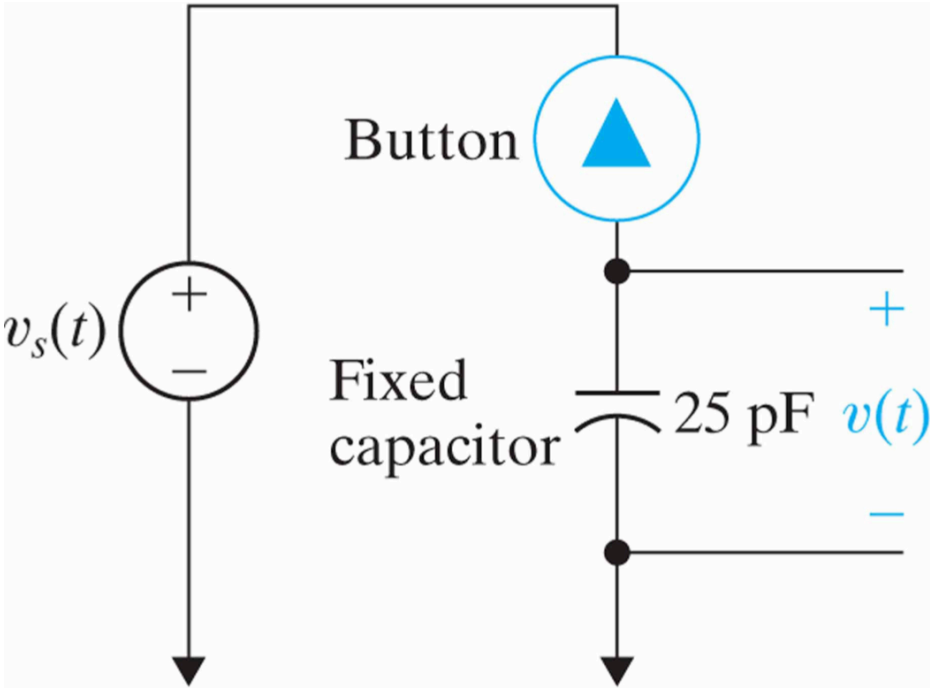
The actual values of the capacitors are in the range of 10 to 50 pF, depending on the *exact geometry of the switch*, *how the finger is inserted*, *whether the person is wearing gloves*, and so forth.

Example #5

Assume that all capacitors have the same value of 25 pF. Also assume the elevator call button is placed in the capacitive equivalent of a voltage-divider circuit, as shown in the Figure.

a) Calculate the output voltage with no finger present.

b) Calculate the output voltage when a finger touches the button.



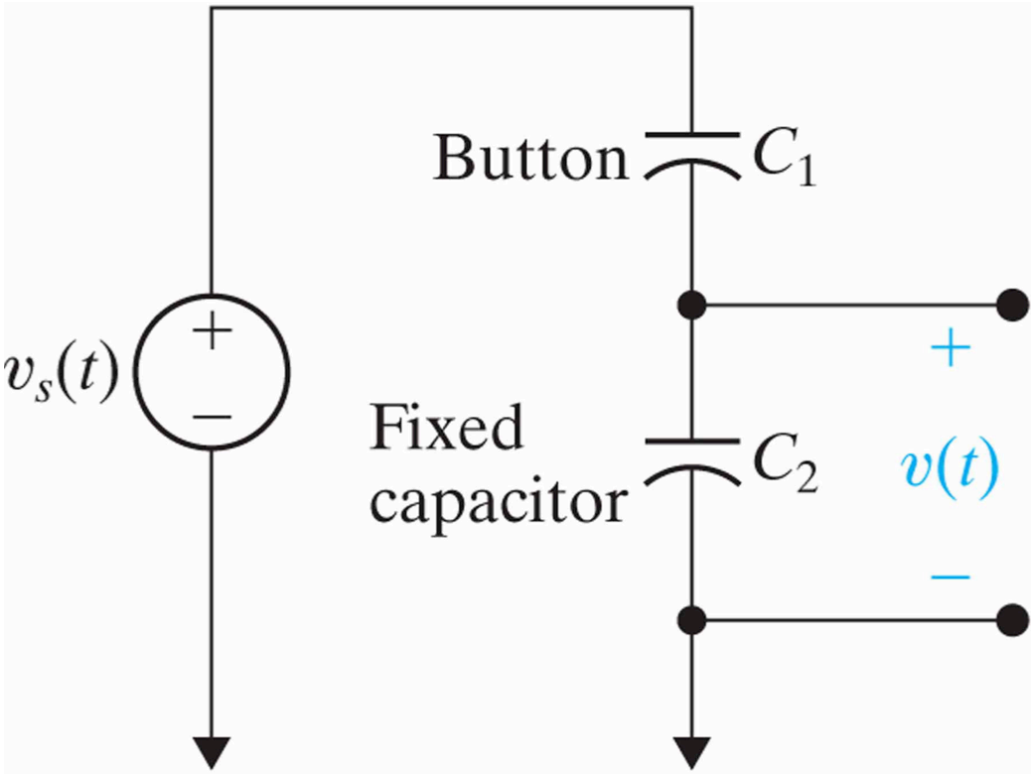
Solution for Example #5

$$\frac{dv}{dt} = \frac{C_1}{C_1 + C_2}\frac{dv_s}{dt}$$

$$C_1\frac{d(v - v_s)}{dt} + C_2\frac{dv}{dt} = 0$$

$$v(t) = 0.5v_s(t)$$

$$v(t) = \frac{C_1}{C_1 + C_2}v_s(t) + v(0)$$

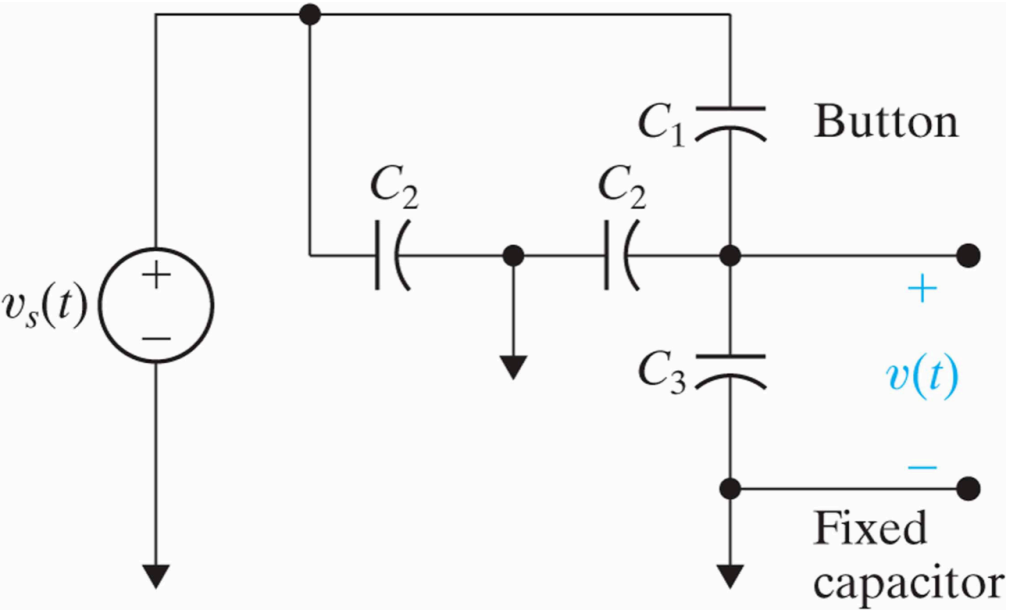


$$C_1\frac{d(v - v_s)}{dt} + C_2\frac{dv}{dt} + C_3\frac{dv}{dt} = 0$$

$$\frac{dv}{dt} = \frac{C_1}{C_1 + C_2 + C_3}\frac{dv_s}{dt}$$

$$v(t) = \frac{C_1}{C_1 + C_2 + C_3}v_s(t) + v(0)$$

$$v(t) = 0.333v_s(t)$$



Summary

- Inductance: *definitions*, *characteristics*, and *equations*
- Capacitance: *definitions*, *characteristics*, and *equations*
- Series-Parallel combinations of Inductance and Capacitance
- Mutual inductance: *dot convention*